

Audit of qml2.0/codes

Scope: current executable code, using the supplied table only as a checklist of computational stages. This is a code audit, not a comparison with that document. No code or datasets were changed. No production calculations, training, or repairs were run.

Findings

1. Default input cannot support the default nine-molecule run - confirmed failure. `run1.py:485,488` selects `results/descriptor` and all nine molecules. Its H2S records retain six AOs, but `circuit.py:125-129` and `run1.ao_selection:140-172` require ten. Loading H2S/001 was reproduced and raises `valence_indices: saved selection does not match canonical valence AO labels at line164`. Thus even with a valid split selection, the default full run stops. The checked descriptor-1s-only H2S record passes. Also, descriptor production defaults to `results/descriptor-v2-methods-t` (`descriptor.py:569`), so producer and consumer defaults do not connect automatically. These are configuration/data-contract findings, not reasons to change the scientific AO selection.
2. `FE_prime` is executable but unavailable in the plotting CLI - confirmed interface gap. Current `run1.py:287,489` accepts `FE_prime`, and `raw_spectra:305-328` computes ascending eigenvalues of the selected Löwdin Fock block. However, `plot_run1_results.py:52,71` restricts `--runs` to FG, FE and MB-1. That plotting entry point cannot admit an `FE_prime` result. This affects output coverage, not the computed spectrum or trained energy.
3. Normal descriptor generation is calculation-only, not scientific acceptance - validation gap. `descriptor.calculate_record:149-157` proceeds from RHF.kernel through canonicalization into RMP2 without rejecting `mf.converged=False` or performing the initial-data stability checks. `lowdin_factors:51-53` applies square roots/inverse square roots without an eigenvalue threshold. `save_record:232-239` records successful return without scientific validation. The producer explicitly stores `validation_performed=False`. The downstream loader rejects unconverged RHF and validates overlap/transforms (`run1.py:191,198-216`), but standalone descriptor completion does not establish scientific validity. No failing-convergence sample was observed in the numerical checks; this is a source-established acceptance gap, not a claimed observed bad result.

Important verified dataflow, not defects: T is calculated and saved but not consumed by current Run 1; MB-1 contains no pair or triple gates. Its pair scale is nevertheless fitted and must be positive. FCI freezes core MOs, while descriptor RMP2 freezes none. AO selection happens after full-space Löwdin transformations. These are actual implementation choices; the audit does not propose changing them.

Computational-stage audit

Source abbreviations: I=initialdata.py; D=descriptor.py; R=run1.py; C=circuit.py; T=train.py; P=plot_run1_results.py, all under /Users/novaz/Desktop/qml2.0/codes.

M denotes full AO/spatial MO count; v_i maps local retained AO i to a full source AO; n denotes the selected descriptor count, or the feature/wire count for geometric fingerprints. Energies are Hartree (Ha), geometry Å. Formulas use ordinary transpose for the implemented real arrays. “Library” marks behavior delegated outside repository code, not an independently audited internal algorithm.

Layer	Exact current source	Verified computation, representation and dataflow
1 Geometry	I optimize_equilibrium771-824; StretchPath194-261; calculate_harmonic431-579; scan_grid858-865	Cartesian RHF optimization; $k=\sum t_{ax} H_{abxy} t_{by}$, $\mu=\sum m_a t_{ax}^2$; harmonic n=1 endpoints $s_e \pm \Delta$, $\Delta=\sqrt{(3\hbar/(\mu\omega))}$ with SI→Å conversion. Inclusive30-point linspace; saved IDs1..30. Raw internal stretch undergoes mass-weighted proper alignment. Output X(N_atom,3); descriptor consumes JSON coordinates.
2 AO overlap	I rhf_arrays887-893; D calculate_record144-161	Molecule geometry/basis/charge/spin and spherical AOs define S(M,M); mf.get_ovlp() delegates integrals to PySCF. Dimensionless overlap; full space retained.
3 RHF reference	I configured_rhf607-613, polish_orbitals616-647, solve_point660-743; D139-159	Initial-data RHF has tight convergence/internal-stability checks. D separately recomputes RHF using saved AO density as its initial guess. $P=C \text{ diag}(\text{occupations}) C^T$; $F=h+J[P]-K[P]/2$. Spin-summed P(M,M), F(M,M) Ha, C(M,M). SCF/integral internals library-bound.
4 Core/space definitions	I ACTIVE_SPACES37-41; D select_valence_aos91-119	FCI frozen MOs and excluded descriptor AOs are independent. MP2/RDM frozen=0. Descriptor exclusion is Z>2 and principal shell1, determined from AO/shell metadata; H2S retains10 AOs.
5 MP2 amplitudes	D calculate_record156-169	RMP2(mf,frozen=0); t2 shape(n_occ,n_occ,n_virt,n_virt), dimensionless, all occupied/virtual spatial MOs. Restricted amplitude/energy construction delegated to PySCF; saved-array amplitude checks agree to roundoff.
6 Used RDM construction	D build_consistent_rdms60-88	Determinant Φ and χ from zero singles and restricted t2; $w=v\text{dot}(\chi,\chi)$. For either RDM, $R1=R[\Phi+\chi]-R[\Phi]-R[\chi]$, $R2=R[\chi]-wR[\Phi]$, $R=R0+R1+R2$. Full MO 1-RDM(M,M), 2-RDM(M,M,M,M). PySCF determinant/RDM APIs are boundaries. Raw MP2 RDMs and exactly normalized state are stored diagnostics.
7 Tensor ordering	D transform_rank456-57; extract_descriptors129-136	Full rank-four arrays retain raw PySCF ordering. Only T extraction applies transpose(0,2,1,3). No explicit four-spin-block pipeline. Raw and permuted indices must remain distinct.
8 MO→AO	D calculate_record171-175; transform_rank456-57	$D_{AO}=C D_{MO} C^T$; $G_{abcd}=\sum_{pqrs} C_{ap} C_{bq} C_{cr} C_{ds} G_{pqrs}$. Full M-sized axes, no conjugation or AO restriction.
9 AO restriction	D122-136,182-186; R ao_selection140-172	Restrict P_L,F_L,Λ_L only after full-space transforms. D stores source-ascending v; R selects canonical atom/AO order. BeH2 model source indices=[5,1,2,3,4,6], not sorted.

Layer	Exact current source	Verified computation, representation and dataflow
10 Disconnected reference	D disconnected_rhf47-48;176	$G0_{pqrs} = P_{pq} P_{rs} - 0.5 P_{ps} P_{rq}$, using full RHF spin-summed P. No summed p,q,r,s in these outer products.
11 Reference subtraction	D176-181	$\Lambda = G_{\text{consistent}} - G0[P_{\text{RHF}}]$, full AO(M^4). The conventional subtraction using correlated density is separately stored; it does not feed descriptor extraction.
12 Extra diagnostic contraction	D160-210	No separate all-ones fourth-axis-sum stage is executed. Stored raw RDMS, order components and alternate cumulant are diagnostics; their existence does not imply model use.
13 FCI and target	I frozen_core_fci896-943; calculate_point946-990; R227-232	Canonical RHF CASCI, active MOs(ncore,M); ncas=M-ncore, Ncas=N_e-2ncore. $E_{\text{FCI}} = E_{\text{active}} + E_{\text{core}}$ including nuclear/core terms. $y = E_{\text{FCI}} - E_{\text{RHF}}$, input. CI shape(choose(ncas,Ncas/2),choose(ncas,Ncas/2)); active h1 and pair-packed h2 saved, not used by D. Solver library-bound.
14 Overlap powers	D lowdin_factors51-53; R208-211	Full $S = U \text{diag}(s) U^T$; $H = U \text{diag}(\sqrt{s}) U^T$, $J = U \text{diag}(1/\sqrt{s}) U^T$. Producer no clipping/threshold; loader requires $\min(s) > 1e-8$ and matrix identities.
15 One-body Löwdin	D182-184; R212-213	$P_L = H P_{\text{RHF}} H$; $F_L = J F_{\text{AO}} J$, both full(M,M). No explicit averaging with transpose here. Correlated density does not supply P_L .
16 Rank-four Löwdin	D56-57,182-184; R216	$\Lambda_{L,abcd} = \sum_{pqrs} H_{ap} H_{bq} H_{cr} H_{ds} \Lambda_{\text{AO},pqrs}$. All four full-space axes transformed before selection.
17 T extraction	D extract_descriptors122-136	$T_{ijk} = \Lambda_L[v_i, v_k, v_j, v_k]$, no sum over k. $T_{\text{full}}(n,n,n)$, selected triples(choose(n,3),). R constructs T=None; current launched models do not consume T.
18 P/F features	D133-136; R218-241	Saved P_{μ}/F_{μ} are selected diagonals(n); P_{munu} strict pairs(choose(n,2),). R validates source order, then rebuilds canonical pii/fii/pij and full selected fij. P dimensionless, F Ha. FE uses eigvalsh(pij); optional FE_prime uses eigvalsh(fij), ascending, classical spectral features.
19 Population	R load_population244-282; retained_positions111-117	Exactly IDs001..030 per molecule, no cutoff filtering. Reject incomplete/error records. Sort by(q_A,geometry_index); zero-based traversal positions.
20 Split	R validate_split_manifest41-73;244-269;331-366	Default explicit disjoint train/validation/test sets cover all30. Explicit15-15 uses odd IDs train/even IDs test; membership is independent of sort order.

Layer	Exact current source	Verified computation, representation and dataflow
21 Diagonal scaling	C compute_encoding_constants345-392; standardize_diagonal394-399	MB scalar mean/std pooled over $N_{\text{train}} \times n$ entries, separately P/F; ddof0. Standardize $(x-\mu)/\sigma$, require positive finite σ ; no clipping/epsilon floor. Training rows only.
22 Pair/triple indexing	C pair_addresses241-244; triple_addresses245-248;249-262	Local $i < j$ row-major and $i < j < k$ nested lexicographic; output lengths choose(n,2),choose(n,3). Pairs feed scale fitting. Triple helper unused in Run1.
23 Pair/triple scales	C _p95_abs279-287;381-392	Linear95th percentile of pooled absolute entries, zeros included; $\alpha = \text{atanh}(.9)/q95$. β only MB-3 helper. Empty/nonpositive inputs reject, no flatten fallback. MB-1 fits α but does not use it in gates.
24 Local encoding	C CachedCircuitBank461-466; encode755-763; FingerprintCircuit800-820	MB $\theta_i = (\pi/2) \tanh(a \text{Pstd}_i + b \text{Fstd}_i + c)$, shared trainable a,b,c. Fingerprints: per-column train mean/std; $\sigma < 1e-12$ masked, safe $\sigma=1$; scale $\pi/2$, zero masks, then clip $[-\pi, \pi]$. RY half-angle convention delegated to PennyLane.
25 Pair encoding	C467-473,610-640	IsingZZ($(\pi/2) \tanh(\alpha P_{ij})$) exists for helper models. MB-1 passes no pair input and queues no pair gates. Not an active Run1 stage.
26 Triple encoding	C474-481	MultiRZ($(\pi/2) \tanh(\beta T_{ijk})$) helper for $i < j < k$. No current Run1 method supplies triple inputs; runtime triple results not verified.
27 HEA	C shared_ansatz416-424	Each layer: same RY(ϕ_i) on all wires, then sequential CNOT $q \rightarrow (q+1) \bmod n$, ascending q, wrap last. L=2, single encoding. n=2 executes both directed edges.
28 Measurement	C482-483,818-820	Ordered Pauli-Z expectation on every physical wire, $z(n)$, dimensionless. Simulation/autodiff library-bound. Independent statevector forward predictions checked; direct runtime z residual not separately obtained.
29 Readout features	C pad_z_to_length568-574; invariant_moments576-583;641-660	$m = \max(8, n)$; zero-pad z on right; $M_k = \text{mean}(z^k)$, $k=1..3$ using unpadded z. $f = [z_{\text{pad}}, M1, M2, M3]$, shape(m+3). No extra qubits.
30 Prediction	C CorrelationEnergyModel655-681; FingerprintModel880-910	$y_{\text{hat}} = b + w \cdot f$, scalar H_a , no target standardization. Fingerprint n is feature width, not necessarily AO count.
31 Initialization	T initialize_parameters18-32; C843-848	Local seeded CPU Torch generator; MB full .05*randn vector, fingerprint separate phi/bias/weight draws; bias overwritten with training target mean. MB $D=L+m+7$, fingerprints $D=L+m+4$.
32 Batches/loss	T SampleOrder34-43; iter_batches45-52; mse_loss70-73;183-233	Separate persistent same-seed shuffle; fresh permutation each epoch; final partial batch retained. $L = \text{mean}((\text{prediction} - \text{target})^2)$ over actual batch, H_a^2 .

Layer	Exact current source	Verified computation, representation and dataflow
33 Adam	T54-91; C850-873	Zero grads to None, backward, model-specific gradient binding, Adam step. MB one vector group; fingerprints shared-storage blocks. Installed Torch defaults and two diagnostic steps checked; complete training trajectory not replayed.
34 Checkpoint	T BestCheckpoint132-173;211-233; R402-453	Strict improvement; earliest tie. Validation MAE for manifest, training MAE for15-15. Restore theta only, not optimizer; final test evaluation after restoration.
35 Metrics	T error_metrics244-254; evaluate256-266	e=prediction-target; MAE=mean(abs(e)); RMSE= $\sqrt{\text{mean}(e^2)}$; max(abs(e)); max(e)-min(e). Scalar Ha and $\times 1000$ mHa after reduction; each split separately.
36 Seed summaries	T compute_statistics268-272; R summarize455-480; P131-149	Require32 seeds in full launcher. Mean/sampleSD(ddof1)/count/median/SE. Overall: equal molecule mean within seed, then seed statistics. Core singleton SD/SE NaN; plot helper SD0,SE NaN.
37 Plot assembly	P validate_seed_runs152-168; load_prediction346-375;377-397	Exact configured seed coverage and equality of sorted(geometry ID,split,bond_length_A,target). Bars test mean $\pm$ SE; curves per-geometry signed-error mean $\pm$ sampleSD. FE_prime CLI gap noted above. Initial-data plot is a separate branch: relative1000(E-E_RHF(re)) or absolute Ha; curve31 points including equilibrium, CSV30 scans.

Cross-stage checks

- The initial RHF/CASCI reference and descriptor RHF recomputation are separate calculations; the loader checks their RHF energies within $1e-10$ Ha. The target remains the input FCI-RHF difference.
- The determinant RDM is truncated through second order; ci_normalized_eta1 and ci_second_order_normalization are stored but do not replace the implemented RDM algebra.
- Full-space matrix powers precede restriction. Stored source-local indices, canonical full-AO indices and circuit wire positions are different index systems.
- FG has explicit molecule-dependent distance/reciprocal/angle/torsion order. H2O2 torsion can be unwrapped across the complete sorted scan before train-only fitting; this is a preprocessing dependency on the full scan, not label usage.
- Current FE_prime uses the Fock spectrum in Ha before per-column scaling. FE uses the density spectrum. Both become dimensionless rotation inputs; neither changes the target definition.
- Historical fixed-angle fields in the molecule registry are unused by the current geometry generator. Historical UHF/UMP2-named sample fields receive None; names do not establish computations.

Numerical verification already completed

Check	Exact shapes / coverage	Maximum absolute residual
Independent aligned geometry reconstruction	27 samples across 9 molecules; (2,3)/(3,3)/(4,3)	5.10702591327572e-15 Å
Harmonic grid reconstruction	9×(30,)	2.220446049250313e-16 Å
FCI-RHF target arithmetic	LiH/H ₂ S/H ₂ O ₂ IDs001/015/030, scalars	0 Ha
Independent LiH active Hamiltonian	H(25,25), CI(5,5); 3 samples	Rayleigh 8.881784197001252e-16 Ha; eigenvector 6.661338147750939e-16 Ha
Full rank-four transforms	LiH/H ₂ S/H ₂ O ₂ ID001; 6 ⁴ /11 ⁴ /12 ⁴ arrays	1.5987211554602254e-14
Literal T indexing against saved current-selection descriptors	(5,5,5)/(10,10,10)/(10,10,10); selected(10)/(120)/(120)	0
Independent NumPy statevector→saved prediction	FG/FE/MB-1, seeds0/31, CO IDs001/016/030; states(4)/(256)/(256), readout(11)	3.3306690738754696e-16 Ha
Saved error/metric recomputation	96 runs; residuals(2880,), metrics(192,8)	0
Earliest training-MAE checkpoint minimum	96 histories×500 epochs; outputs(96,)	0
Independent seed summary	(6,5)	4.440892098500626e-16

These samples support the checked arithmetic, not universal correctness. PySCF's compiled integral/SCF/determinant engines, PennyLane derivatives, and a full training replay were not independently audited. The saved learning run is CO-only and 15-15; validation-selected training, multi-molecule aggregation at runtime, FE_prime training and MB-2/3 runtime remain NOT_VERIFIABLE from these numerical samples. Their described code paths were inspected.

Two files, run1.py and circuit.py, changed externally during this session to add FE_prime. Final source references above use the refreshed files; earlier numerical runtime checks are identified as pre-addition evidence, not a blanket certification of the new branch. Independent saved-array/statevector arithmetic remains valid for the named saved samples. evidence.json records earlier and final source hashes and only code/numerical evidence; no document-comparison results are included.